

Build Your Own Private Cloud with EUCALYPTUS

Now empower your organisation the open source way! Learn about the Eucalyptus private cloud platform and then go on to install, configure and work with it.

Eucalyptus is an open source cloud platform that enables organisations to create private clouds inside their data centres using existing virtualised infrastructure. First released in 2008, 'Eucalyptus' is actually an acronym that stands for 'Elastic Utility Computing Architecture for Linking Your Programs to Useful Systems'.

Eucalyptus enables IT organisations to build Amazon Web Services (AWS)-compatible private clouds that can pool together existing virtualised IT resources and provide them to its customers in a flexible, on-demand, pay-as-you-go basis.

The Eucalyptus cloud platform primarily comprises of five main components, each briefly explained below:

Cloud controller (CLC): The cloud controller is essentially a Web-based interface between Eucalyptus and the outside world. It provides cloud administrators an interface with which they can configure and manage the cloud's underlying compute, storage and network resources. It also handles high-level user authentication, quota management along with a few basic accounting and reporting mechanisms. Users can also query the CLC using Eucalyptus's command line tools called Euca2ools.

Walrus: Walrus provides persistent storage capabilities to all of the virtual machines in the Eucalyptus cloud. It is basically a large storage container where users can upload data from any file type using simple HTTP *put-get* queries.

Cluster controller (CC): A typical Eucalyptus cloud set-up can comprise multiple nodes, which are actually hypervisors on which virtual machines are provisioned. These nodes can be collectively grouped and managed by using the cluster controller, which primarily looks after the inter-node communications and Service Level Agreements (SLAs) of each cluster.

Storage controller (SC): The storage controller provides persistent block-level storage for virtual machines that are launched within a Eucalyptus cloud. The SC interfaces with a wide variety of storage systems including local file systems, NFS, SAN and even iSCSI.

Node controller (NC): Node controllers or nodes are hypervisor-based virtualised servers that host virtual



machines which are provisioned by the Eucalyptus cloud. Eucalyptus additionally provides an optional add-in component that even supports VMware ESXi hypervisor as a node, called a VMware broker.

Features

The latest release of Eucalyptus (Version 3.3.2) introduces many new features, some of which are explained below.

Auto-scaling: Mirrored on the concept of auto-scaling by AWS, Eucalyptus too provides similar APIs that help cloud administrators set rules and policies that enable applications running on virtual machines to scale-up and scale-down dynamically, based on fluctuating workloads.

Elastic load balancing: This is basically a service that provides better availability and scalability for applications running atop the Eucalyptus cloud. It automatically senses and distributes incoming application traffic and service calls across multiple running instances, thus improving the performance and uptime of the application.

CloudWatch: This provides monitoring for virtual machines and applications hosted on the Eucalyptus cloud. Using CloudWatch, users can programmatically track, collect and analyse metrics, set alarms to troubleshoot performance issues and take automated action as well, based on the state of the cloud environment.